

Student Exam Performance Analysis

An Exploratory Data Analysis Report

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Python • Pandas • Matplotlib • Seaborn • Jupyter Notebook

1. Objective

This report examines a dataset of student exam scores to understand which factors are most strongly associated with academic performance. The analysis investigates the relationship between exam outcomes and test preparation, parental education level, lunch type (a proxy for economic background), race/ethnicity group, and gender, with the goal of surfacing patterns that are both statistically clear and practically meaningful.

2. Dataset Overview

The dataset used is the publicly available **Students Performance in Exams** dataset (Kaggle), consisting of 1,000 student records across 8 columns. It contains no missing values and no duplicate rows.

Column	Description
gender	Student's gender (male / female)
race/ethnicity	Categorical group (group A through group E)
parental level of education	Highest education level attained by a parent
lunch	Lunch type (standard / free-reduced) - a proxy for economic background
test preparation course	Whether the student completed a test prep course
math score	Score out of 100
reading score	Score out of 100
writing score	Score out of 100

3. Methodology

- Loaded and inspected the dataset (shape, column types, null and duplicate checks).
- Cleaned column names to a consistent lowercase, underscore-separated format.
- Engineered a new feature, **average_score**, calculated as the mean of math, reading, and writing scores.
- Conducted univariate analysis on the overall score distribution.
- Conducted bivariate analysis comparing average_score across test preparation, parental education, lunch type, and race/ethnicity.
- Conducted two-factor analysis combining lunch type and test preparation to uncover a deeper, non-obvious pattern.
- Analyzed the correlation between math, reading, and writing scores using a heatmap.

4. Analysis and Findings

4.1 Overall Score Distribution

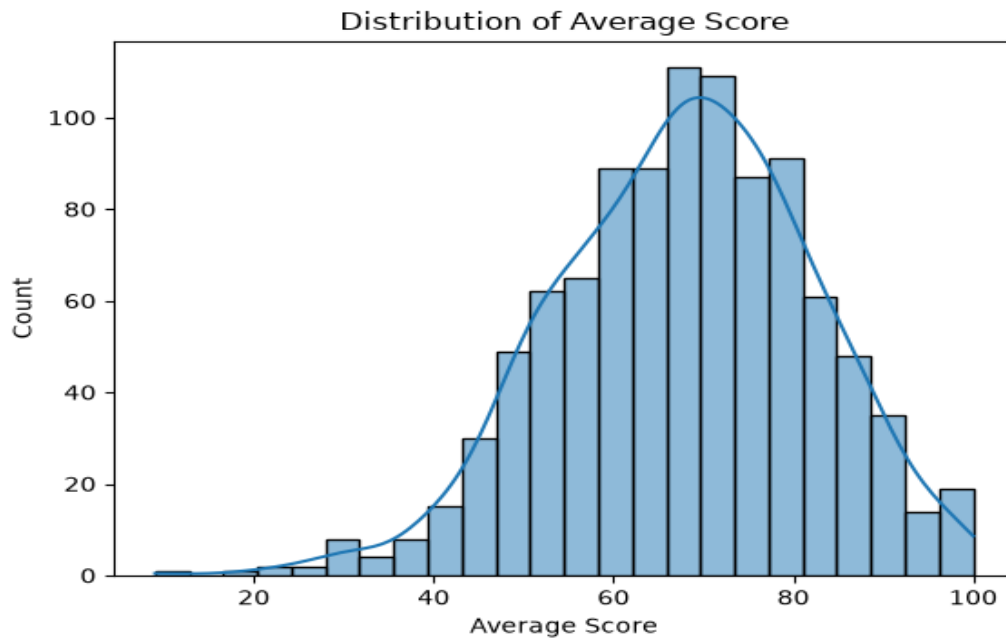


Figure 1: Histogram of average_score with KDE overlay.

- The distribution of average_score is left-skewed, meaning most students scored above the average, while a smaller group of lower-scoring students pulls the tail of the distribution to the left.

4.2 Test Preparation Course

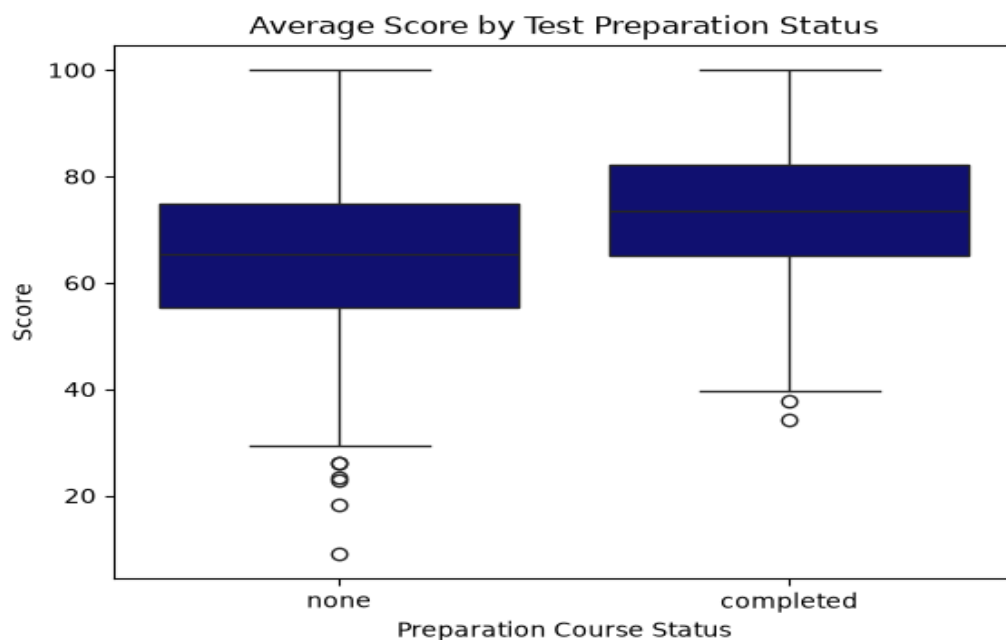


Figure 2: Average score by test preparation course status.

- Students who completed the test preparation course averaged 72.67, compared to 65.04 for those who did not - a gap of approximately 7.6 points.
- This suggests test preparation has a meaningful, measurable impact on exam performance.

4.3 Parental Level of Education

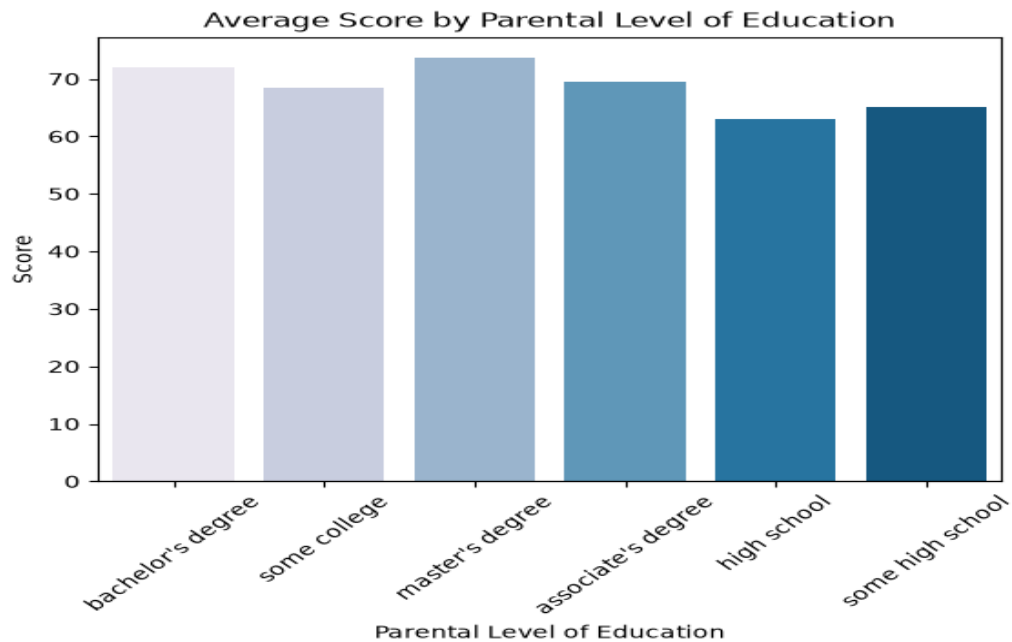


Figure 3: Average score by parental level of education.

- Average score rises fairly steadily with parental education, from 65.11 (some high school) up to 73.60 (master's degree).
- One irregularity: students whose parents completed high school (63.10) scored slightly lower than those whose parents had only some high school education (65.11) - a small deviation from the otherwise increasing trend.

4.4 Lunch Type

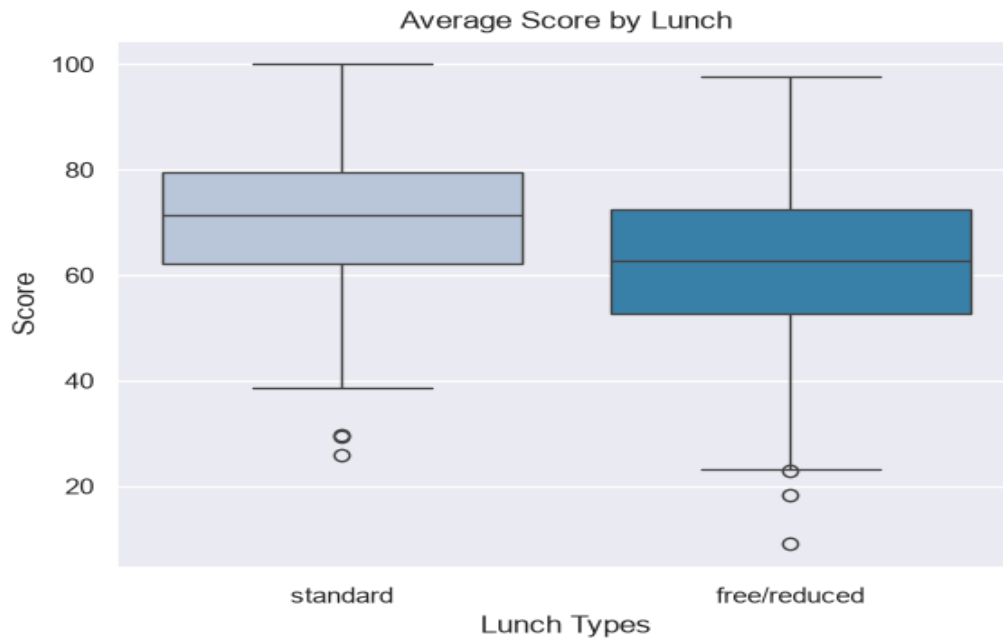


Figure 4: Average score by lunch type.

- Students with standard lunch averaged 70.84, compared to 62.20 for free/reduced lunch - a gap of approximately 8.6 points.
- This is the largest single-factor gap identified in the dataset, and likely reflects broader socioeconomic differences tied to lunch assistance eligibility.

4.5 Lunch Type and Test Preparation Combined

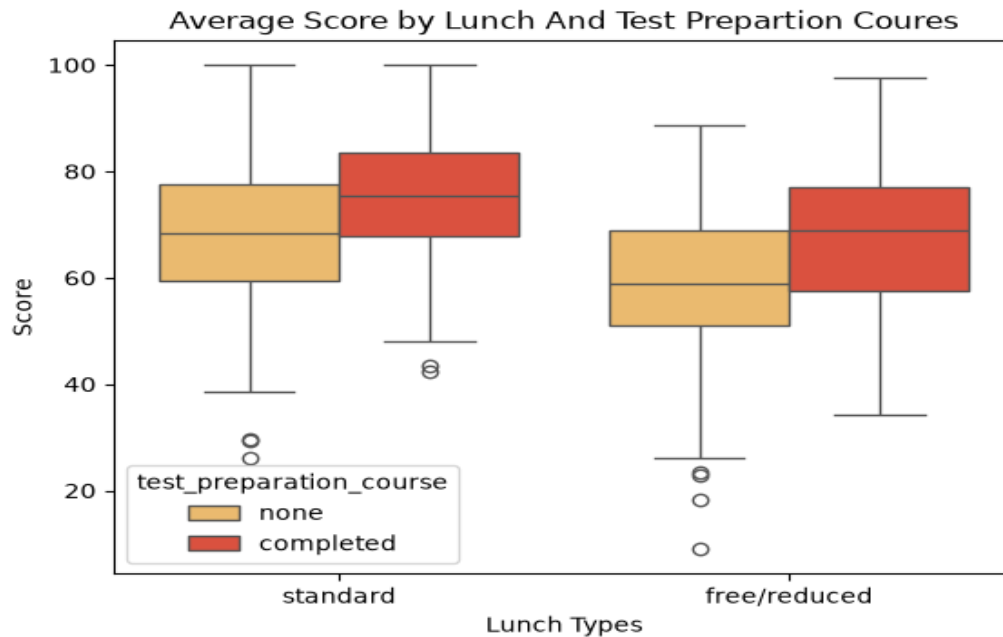


Figure 5: Average score by lunch type, split by test preparation status.

- Students with free/reduced lunch who did not complete test prep scored lowest on average (58.95), while students with standard lunch who completed test prep scored highest (75.51) - a gap of 16.5 points between the extremes.
- Test preparation raised scores for free/reduced lunch students by approximately 8.8 points, slightly more than the 7.2 point boost seen for standard lunch students.
- This suggests test preparation may help narrow, rather than widen, the performance gap linked to economic background.

4.6 Race/Ethnicity

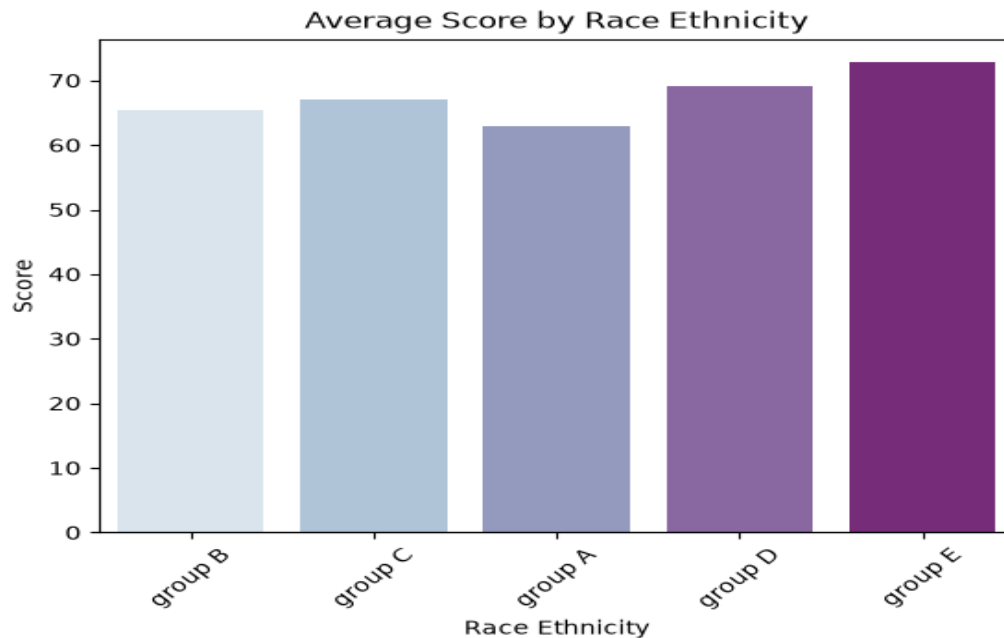


Figure 6: Average score by race/ethnicity group.

- Average score increases consistently from group A (62.99) to group E (72.75), a spread of approximately 9.8 points.
- Unlike some other factors, this increase is gradual and consistent across all groups rather than concentrated in just one or two categories.

4.7 Correlation Between Subject Scores

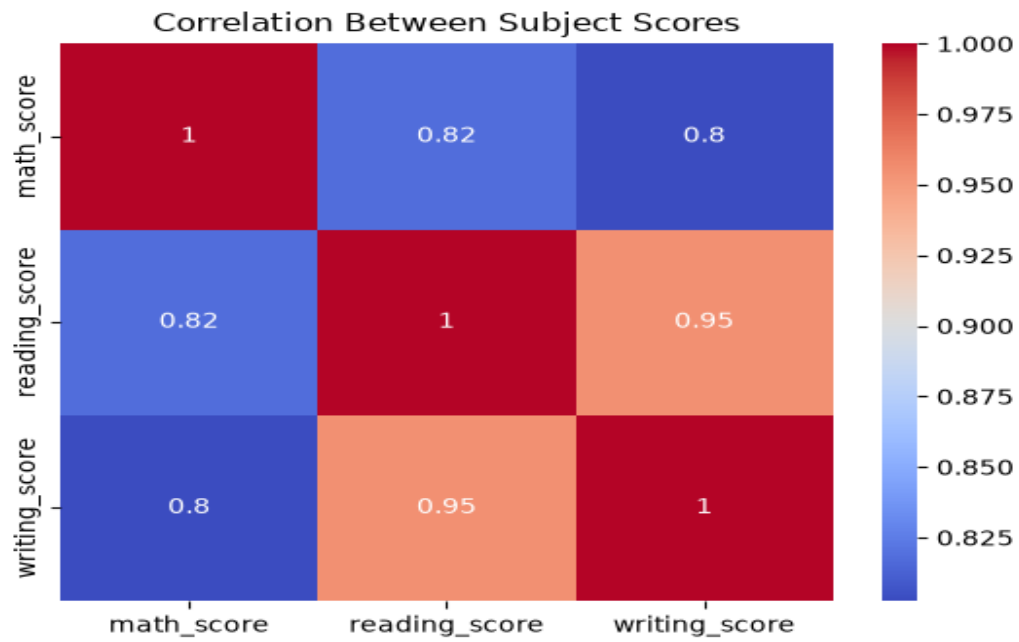


Figure 7: Correlation heatmap between math, reading, and writing scores.

- Reading and writing scores are highly correlated (0.95), meaning students who perform well in one nearly always perform well in the other.
- Math shows a comparatively weaker relationship with both reading (0.82) and writing (0.80), suggesting math ability is somewhat more independent of general language performance.

5. Key Insights Summary

- **Test preparation matters** - a consistent ~7.6 point improvement in average score.
- **Economic background (lunch type) shows the largest single-factor gap** - approximately 8.6 points.
- **Test preparation appears to narrow the economic performance gap** rather than widen it, benefiting free/reduced lunch students slightly more than standard lunch students.
- **Parental education shows a clear step-up pattern**, with an 8.5 point spread from the lowest to highest education level.
- **Race/ethnicity groups show a steady, gradual increase** in average score, a spread of about 9.8 points.
- **Reading and writing are nearly interchangeable** (correlation of 0.95), while math is comparatively more independent of language performance.

6. Future Improvements

- Explore whether gender interacts with subject-specific performance (e.g. does the gender gap differ between math and writing?).
- Build a simple regression model to quantify which factor is the strongest predictor of average score.
- Expand the dataset with additional features (e.g. study hours, attendance) if a larger version becomes available.

Dataset source: Students Performance in Exams, Kaggle ([spscientist/students-performance-in-exams](https://www.kaggle.com/spscientist/students-performance-in-exams)).